CD705: Exercises, Week 6

11 November 2026

Exercise 1. The film *Harry Potter and the Sorcerer's Stone* broke the U.S. box office debut record previously held by *The Lost World: Jurassic Park* according to The Wall Street Journal, November 19, 2001.

The Lost World: Jurassic Park took in \$72.1 million in its first three-day weekend.

Harry Potter and the Sorcerer's Stone was shown in 3672 cinemas. A sample of 100 cinemas showed that the mean three-day weekend gross was \$25 467 per cinema and the sample standard deviation was \$4 980.

a) Find a 95% confidence interval for the mean three-day weekend gross per cinema.

b) Calculate an interval estimate of the total *Harry Potter and the Sorcerer's Stone* took in during its first three-day weekend.

c) Does you result agree with the claim that *Harry Potter and the Sorcerer's Stone* broke the U.S. box office debut record?

Exercise 2. The following are head circumferences (in centimetres) at birth of 15 infants (see `heads.csv` on My Studies):

33.38	32.15	33.99	34.10	33.97
34.34	33.95	33.85	34.23	32.73
33.46	34.13	34.45	34.19	34.05

Is it reasonable to conclude that the population mean head circumference is different from 34.2 cm?

Exercise 3. A machine should fill sugar cartons nominally containing 1 kg, with the standard deviation of 0.03 kg.

It was thought that the machine was overfilling the cartons, and 11 cartons of sugar were randomly selected from a large batch. The weights of sugar were (see `sugar.csv` on My Studies): 1.02, 1.05, 1.08, 1.03, 1.00, 1.06, 1.08, 1.01, 1.04, 1.07, 1.00 (in kg).

Calculate appropriate confidence interval(s). Formulate an appropriate hypothesis and test it. State your conclusion.

Exercise 4. For Exercise 19, Week 5, formulate an appropriate hypothesis, test it and state your conclusion.

Exercise 5. (cf. Exercise 2, Week 3)

The manufacturer claims that the median lifetime of a component is 25 hours (that means that the lifetime of such a component is equally likely to be below and above 25 hours). To verify this claim, an analyst decided to select a random sample of 20 such components, observe their lifetime and count X, the number of observations exceeding 25 hours.

a) The analyst decided to reject the manufacturer's claim if $x \leq 4$ or $x \geq 16$. Calculate the probability of Type I error for this procedure.

b) If we want to reject if $x \leq \delta$ or $x \geq 20 - \delta$ and the required significance level is 5%, what value of δ should we select?

c) If the manufacturer claims that the median lifetime of the component is *at least* 25 hours, devise an appropriate testing procedure.

Exercise 6. 12 basketball players were told to shoot 25 free throws and the number of goals was recorded. They were then given an extensive workout and then told to shoot another 25 free throws; the number of goals was again recorded. The data are in the file `basketball.sav`.

Does a workout affect goal scoring ability?

Exercise 7. Maximum total phosphorus (mg/l) was determined in samples of water taken monthly from two sites in Chesapeake Bay from October to September. The results are in the file `phosphorus.csv`.

Is there a significant difference in the level of maximum total phosphorus between the two sites?

Exercise 8. As part of a study into lead pollution and its absorbance by seaweed, two species of fucoid seaweed, spiralis and serratus were investigated. The purpose of the investigation was to assess whether lead levels differ between the two species. Samples from 10 plants at each location were taken and assessed for levels of lead, measured as microgram lead to gram dry weight. The results are in the file `seaweed.csv`.

Is there evidence of a difference between the two species in the mean lead level?

Exercise 9. It is generally accepted that gender differences in response to heat stress do exist. A group of 10 men and 8 women were put through a vigorous exercise programme that involved the use of a treadmill. The environment was hot, and a minimal amount of water was available to the subjects. The variable of interest is the percentage of body weight lost. The obtained data are in the file `heatstress.csv`.

Do men and women respond differently to heat stress?